LUCAS MOUGEOT

Senior Backend Engineer | High-Performance Data Systems

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SUMMARY

Backend engineer specializing in deterministic, scalable systems. Experienced designing high-integrity Python services, PostgreSQL-backed data models, and containerized multi-service architectures. Comfortable owning system architecture end-to-end with a focus on correctness, concurrency, performance, and operational reliability.

EXPERIENCE

Integration Engineer — EZ Systems (Full-Time) Feb 2025 – Aug 2025

- Architected a unified QA and deployment framework integrating AI inference services, sensor validation pipelines, and CI/CD workflows, ensuring reproducible releases and reducing pre-deployment validation cycles.
- Led software architecture and validation workflows for an autonomous mobile platform operating in production pilot environments, focusing on reliability, fault isolation, and deterministic system behavior.
- Built structured logging, diagnostics, and validation tooling to support repeatable testing, performance analysis, and controlled rollouts across heterogeneous system components.
- Authored technical architecture documents and internal engineering standards guiding reproducible AI-integrated system design.

Founding Systems Engineer — Stealth Platform 2024 – Present

- Built and operate a production-grade backend platform (Python, TypeScript, PostgreSQL) supporting structured intake workflows, rule-based evaluation services, and background processing pipelines.
- Designed a deterministic Python ingestion engine compiling arbitrary repositories into immutable structural and dependency graphs via AST-based analysis, enabling reproducible normalization, validation, and downstream analytics.
- Implemented batch and event-driven processing pipelines handling ordered, versioned transformations with strict idempotency, concurrency, and correctness guarantees.
- Enforced schema-first system boundaries isolating probabilistic AI components behind deterministic validation layers, ensuring replayable state transitions and audit-grade traceability.
- Designed versioned knowledge graphs in PostgreSQL (ltree) with temporal validity windows and override chains to support historical reconstruction and cross-service consistency guarantees.
- Implemented batch and event-driven processing pipelines handling ordered, versioned transformations with strict idempotency and correctness constraints.
- Containerized a multi-service architecture (PostgreSQL, Redis, BullMQ) and built regression, fuzz, and concurrency test suites to guarantee correctness under parallel workloads and high-concurrency execution.

Graduate Researcher — Materials Science (NASA-Funded Capillary Devices) 2023 – 2024

- Designed reproducible experimental workflows across 36 controlled formulations, optimizing performance metrics through structured iteration and statistical evaluation.
- Built MATLAB-based computer-vision pipelines for segmentation, morphology quantification, and ranked candidate selection under controlled experimental constraints.

Undergraduate Researcher — Computational Geometry (NSF-Funded) 2020 – 2022

- Implemented performance-critical computational geometry algorithms in modern C++ (CGAL) under strict planarity and stretch-factor constraints, contributing to a peer-reviewed ACM publication.
- Designed locality-aware spatial algorithms for dense, heterogeneous datasets with constrained graph topology guarantees.

SELECTED PROJECTS

Automated Lithium-Ion Battery Pack Assembler Senior Design Capstone

- Designed and implemented a deterministic automation system using PLC-based ladder logic coordinating motion sequencing, interlocks, and fault-handling state transitions.

- Built repeatable I/O mapping, validation workflows, and controlled bring-up procedures ensuring predictable system behavior under hardware constraints.

EDUCATION

University of North Florida

M.S. Materials Science & Engineering (Thesis-Based) — GPA: 4.00 / 4.00

- Thesis: Development and Characterization of Ceramic Devices for Capillary Action in Microgravity for the Purpose of Growing Vegetables in Space (NASA-Funded)

B.S. Electrical Engineering — GPA: 3.93 / 4.00

B.A. Interdisciplinary Studies: Morphology of Futurism — GPA: 4.00 / 4.00

Concentrations: Mechanical Engineering, Materials Science, Chemistry, Physics, Computing, Mathematics, Mandarin Chinese

PUBLICATIONS & PRESENTATIONS

- Bounded-Degree Plane Geometric Spanners in Practice — ACM Journal of Experimental Algorithmics (Peer-Reviewed)
- Highly Sensitive and Stable pH Sensor Electrodes of TiN Fabricated Using HiPIMS with Kick — AVS 69 Poster, 2023
- Interactive Tool for Bounded-Degree Plane Geometric Spanners — SoCG Media Exposition, 2021

TECHNICAL SKILLS

Languages: Python, C++, SQL, TypeScript, JavaScript
Infrastructure & Systems: Docker, PostgreSQL, Redis, containerized service architectures, CI/CD pipelines, concurrency-safe design, regression and fuzz testing
Backend & Data: API design, background workers, event-driven processing, schema design, distributed systems fundamentals
Tools: GitHub Actions, Supabase, BullMQ, Playwright